

Term Project
Performance Analysis of Turbine, Compressor, and Nozzle
Applications

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Department of Mechanical Engineering

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1 Abstract

In this project three steady flow devices i.e. turbine, compressor and nozzle were studied. Different fluid was used for each flow device. Steam for turbine, refrigerant 134a for compressor and air for nozzle. The aim of project is to study the effect of isentropic efficiency on power output for turbine, power input for compressor and exit velocity for nozzle. For each application seven values of isentropic efficiency were chosen covering reasonable range of efficiencies. A reasonable value for inlet conditions and exit pressure were assumed. Power output for turbine, power input for compressor and exit speed for nozzle was plotted against isentropic efficiency. Also, mass flow rate, exit momentum was considered for actual and isentropic case. The analysis was done on both EES software and manually to ensure comprehensive examination of results. Results from both methods were consistent and valid. It was found that increase in isentropic efficiency increases power output from turbine, decreases power input for compressor and increase exit speed of nozzle. So higher isentropic efficiency is desirable for steady flow devices.

2 Introduction

The efficiency and performance of turbomachinery components, such as turbines, compressors, and nozzles, have significant importance in a range of technical applications within the fields of fluid dynamics and thermodynamics [1]. The present study focuses on doing a comprehensive analysis of the individual components involved in this project, wherein each component operates using a specific fluid. Specifically, the turbine utilises steam as its working fluid, the compressor employs Refrigerant 134a, and the nozzle treats air as an ideal gas.

The main aim of this study is to examine the influence of isentropic efficiency on the operational features of these applications. In order to accomplish this objective, a thorough array of activities has been delineated for each individual component. The inlet conditions and exit pressures are well defined, serving as a fundamental basis for later research.

The research encompasses three primary elements, namely turbines and compressors. The primary objective is to compute power for different isentropic efficiency values. Additionally, the investigation also includes an analysis of nozzles, where both actual and ideal isentropic exit velocities are evaluated. The scope of the inquiry encompasses an examination of the impact of mass flow rate on the momentum at the nozzle exit, hence providing a comprehensive comprehension of performance measures.

In addition, the use of computational tools, such as the Engineering Equation Solver (EES) software, alongside a manual calculation methodology, guarantees a comprehensive and reliable analysis of the outcomes. The juxtaposition of these two methodologies assists to authenticate the precision and dependability of the results, offering a robust groundwork for subsequent investigations.

The primary objective of this research is to analyse and understand the complex interactions among these components. Additionally, it seeks to offer significant findings that have practical consequences for enhancing the performance of turbines, compressors, and nozzles in engineering applications. The anticipated outcomes of this study are expected to make a substantial contribution to the ongoing academic conversation in the fields of fluid dynamics and thermodynamics. These discoveries will provide valuable guidance for future developments in the design and operation of turbomachinery systems.

3 Problem Statement

Three applications need to be considered in the problem i.e. Turbine, compressor and Nozzle.

Fluid for each application was specified as. Steam for turbine, Refrigerant 134a for compressor and Air (Assumed as ideal gas) for nozzle.

Select reasonable conditions for inlet of each case and pressure value at exit for analysis.

For each of three applications, select at least five values of the isentropic efficiency. Make sure to select a good range of the efficiency values with a good interval between them.

Following tasks were need to be completed for each application.

For the compressor and the turbine: Calculate the power for each of the efficiency values and plot the power vs efficiency.

For Nozzle: Calculate the actual and ideal isentropic (maximum) exit speed. Plot the velocity vs nozzle efficiency. S

Present calculations of EES software and manual method. Also, show that similar results were generated by both of these approaches.

4 Solution

Considering following values of isentropic efficiency for three applications. 60%, 65%, 70%, 75%, 80%, 85%, 90%. Two extensive properties were required to specify state of pure substance (single component and single phase) So, for inlet conditions temperature and

pressure were specified in Table: 1 below for each application and only pressure at outlet was specified, temperature will be calculated by using assumed isentropic efficiency. Also, velocity at inlet of turbine, compressor and nozzle was assumed to be negligible. Assume mass flow rate of 3 kg/s for nozzle.

Table 1: Inlet and outlet conditions for each device

Application	Fluid	Inlet Conditions		Outlet Conditions
		Pressure (P ₁)	Temperature (T ₁)	Pressure (P ₂)
		kPa	°C	kPa
Turbine	Steam	12500	500	50
Compressor	R-134a	300	30	1500
Nozzle	Air	200	677	110

Now Required tasks were completed for each application separately.

4.1 Turbine

General definition of isentropic efficiency for steam turbine.

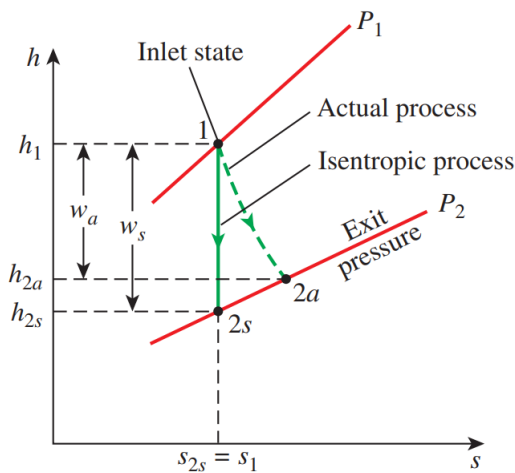


Figure 1: The h-s diagram of the actual and isentropic process of a turbine

Therefore, formula for isentropic efficiency of turbine become [2]

$$\eta_T = \frac{\text{Actual turbine work}}{\text{Isentropic turbine work}} = \frac{w_a}{w_s} = \frac{h_1 - h_{2a}}{h_1 - h_{2s}} \dots \dots \dots eq(1)$$

Specific power output from turbine will be given by

$$w_a = h_1 - h_{2a} \dots \dots \dots eq(2)$$

4.1.1 Manual Calculations

Sample calculation for isentropic efficiency of 60% and given inlet and outlet conditions.

Determine enthalpy and entropy at inlet of turbine from superheated steam tables.

$$h_1 = 3343.6 \text{ kJ/kg} \quad , \quad s_1 = 6.4651 \text{ kJ/kg.K}$$

Now for h_{2s} consider isentropic process from state 1 to state 2 i.e. $s_2 = s_1$

Now at pressure P_2 look for entropy s_2 in steam table [2].

$$s_2 < s_{g@P_2}$$

So, state 2 is saturated liquid vapour mixture. From steam table

$$s_g = 7.5931 \text{ kJ/kg.K} \quad , \quad s_f = 1.0912 \text{ kJ/kg.K}$$

$$h_f = 340.54 \text{ kJ/kg} \quad , \quad h_g = 2645.2 \text{ kJ/kg}$$

$$x_2 = \frac{s_2 - s_f}{s_{fg}} = \frac{6.4651 - 1.0912}{6.5019} = 0.8265$$

$$h_{2s} = h_f + x_2 h_{fg} = 2245.4 \text{ kJ/kg} \quad , \quad T_{2s} = 81.32^\circ\text{C}$$

So, from definition of isentropic efficiency i.e. eq (1)

$$0.6 = \frac{3343.6 - h_{2a}}{3343.6 - 2245.4}$$

Solve above equation for h_{2a}

$$h_{2a} = 2684.7 \text{ kJ/kg}$$

So, power output from turbine from eq (2)

$$w_a = 3343.6 - 2684.7 = 658.92 \text{ kJ/kg}$$

Power output for different efficiency values were given in table below.

Table 2: Power output for different turbine isentropic efficiencies

Isentropic Efficiency	h_1	s_1	h_{2s}	h_{2a}	Power Output
%	kJ/kg	kJ/kg.K	kJ/kg	kJ/kg	kJ/kg
60	3343.6	6.465	2245.4	2684.68	658.92
65	3343.6	6.465	2245.4	2629.77	713.83
70	3343.6	6.465	2245.4	2574.86	768.74
75	3343.6	6.465	2245.4	2519.95	823.65
80	3343.6	6.465	2245.4	2465.04	878.56
85	3343.6	6.465	2245.4	2410.13	933.47
90	3343.6	6.465	2245.4	2355.22	988.38

4.1.2 EES Calculations

Build in property functions were used for calculation in EES Following section display equations solved on EES and table show solution for power output at different isentropic efficiency of turbine.

EES Code	Formatted Equations
"inlet consitions"	inlet consitions
P_1= 12500 [kPa]	$P_1 = 12500 \text{ [kPa]}$
T_1=500 [C]	$T_1 = 500 \text{ [C]}$
"outlet pressure"	outlet pressure
P_2= 50 [kPa]	$P_2 = 50 \text{ [kPa]}$
h_1=	
Enthalpy(Water,T=T_1,P=P_1)	$h_1 = h(\text{water}, T=T_1, P=P_1)$
s_1=Entropy(Water,T=T_1,P=P_1)	$s_1 = s(\text{water}, T=T_1, P=P_1)$
h_2s=Enthalpy(Water,P=P_2,s=s_1)	$h_{2s} = h(\text{water}, P=P_2, s=s_1)$
"Isentropic Efficiency "	Isentropic Efficiency
eta_T= 0.6	$\eta_T = 0.6$
h_2a=h_1 - eta_T*(h_1-h_2s)	$h_{2a} = h_1 - \eta_T \cdot (h_1 - h_{2s})$
"work output"	work output
w_a= h_1-h_2a	$w_a = h_1 - h_{2a}$

Table 3: Power output for different turbine isentropic efficiencies from EES

	1	2	3	4	5	6	7
	η_T	h_1	s_1	h_{2s}	h_{2a}	w_a	
Run 1	0.6	3342	6.462	2245	2684	658.6	
Run 2	0.65	3342	6.462	2245	2629	713.5	
Run 3	0.7	3342	6.462	2245	2574	768.3	
Run 4	0.75	3342	6.462	2245	2519	823.2	
Run 5	0.8	3342	6.462	2245	2464	878.1	
Run 6	0.85	3342	6.462	2245	2409	933	
Run 7	0.9	3342	6.462	2245	2354	987.9	

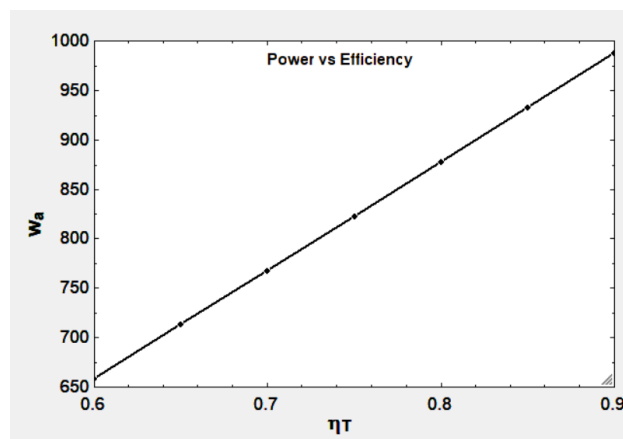


Figure 2: Power vs efficiency plot for turbine

4.2 Compressor

General definition of isentropic efficiency for compressor

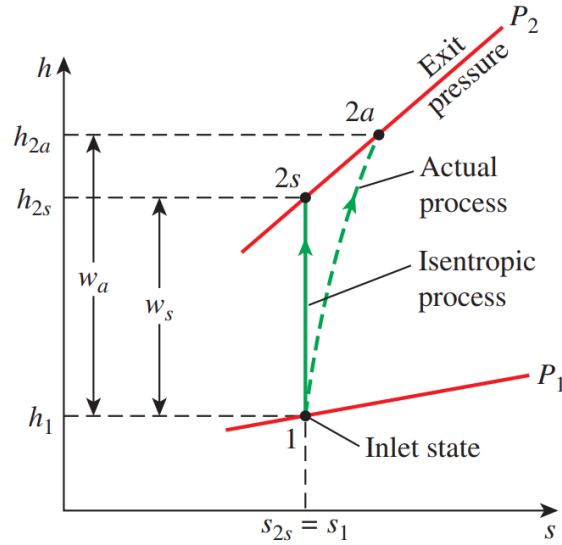


Figure 3: The h - s diagram of the actual and isentropic process of a compressor

Therefore, formula for isentropic efficiency of compressor will be [2]

$$\eta_c = \frac{\text{Isentropic compressor work}}{\text{Actual compressor work}} = \frac{w_s}{w_a} = \frac{h_{2s} - h_1}{h_{2a} - h_1} \dots \dots \dots eq(3)$$

Power input to compressor is given by the following equation

$$w_a = h_{2a} - h_1 \dots \dots \dots eq(4)$$

4.2.1 Manual Calculations

Sample calculation for isentropic efficiency of 60% and given inlet and outlet conditions.

At given inlet conditions R-134a is superheated. From superheated property tables enthalpy and entropy at state 1 is

$$h_1 = 277.03 \text{ kJ/kg} \quad , \quad s_1 = 1.0221 \text{ kJ/kg.K}$$

Now for h_{2s} consider isentropic process from state 1 to state 2 i.e. $s_2 = s_1$

Now at pressure P_2 look for entropy s_2 in property table.

$$s_2 > s_{g@P_2}$$

So, state 2 is also superheated. From superheated table

$$h_{2s} = 317.945 \text{ kJ/kg} \quad , \quad T_{2s} = 90^\circ\text{C}$$

So, from definition of isentropic efficiency i.e. eq (3)

$$0.6 = \frac{317.945 - 277.03}{h_{2a} - 277.03}$$

Solve above equation for h_{2a}

$$h_{2a} = 345.22 \text{ kJ/kg}$$

So, power output from turbine from eq (2)

$$w_a = 345.22 - 277.03 = 68.19 \text{ kJ/kg}$$

Power output for different efficiency values were given in table below.

Table 4: power input for different isentropic efficiencies of compressor

Isentropic Efficiency	h_1	s_1	h_{2s}	h_{2a}	Power Input
%	kJ/kg	kJ/kg. K	kJ/kg	kJ/kg	kJ/kg
60	277.03	1.022	317.945	345.22	68.19
65	277.03	1.022	317.945	340.5	63.47
70	277.03	1.022	317.945	336.3	59.27
75	277.03	1.022	317.945	332.9	55.87
80	277.03	1.022	317.945	328.8	51.77
85	277.03	1.022	317.945	326.2	49.17
90	277.03	1.022	317.945	322.1	45.07

4.2.2 EES Calculations

"inlet consitions" P_1= 300 [kPa] T_1=30 [C] "outlet pressure" P_2= 1500 [kPa] h_1= Enthalpy(R134a,T=T_1,P=P_1) s_1=Entropy(R134a,T=T_1,P=P_1) h_2s=Enthalpy(R134a,P=P_2,s=s_1) "Isentropic Efficiency " eta_T= 0.6 h_2a=h_1+(h_2s - h_1)/eta_T "work input" w_a= h_2a-h_1	Formatted Equations inlet consitions P₁ = 300 [kPa] T₁ = 30 [C] outlet pressure P₂ = 1500 [kPa] h₁ = h (R134a , T=T₁ , P=P₁) s₁ = s (R134a , T=T₁ , P=P₁) h_{2s} = h (R134a , P=P₂ , s=s₁) Isentropic Efficiency η_T = 0.6 $h_{2a} = h_1 + \frac{h_{2s} - h_1}{\eta_T}$ work input w_a = h_{2a} - h₁
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Table 5: power input for different isentropic efficiencies of compressor from EES

1..7	1 η_T	2 h_1	3 s_1	4 h_{2s}	5 h_{2a}	6 w_a
Run 1	0.6	277	1.022	315.8	341.7	64.7
Run 2	0.65	277	1.022	315.8	336.7	59.72
Run 3	0.7	277	1.022	315.8	332.5	55.45
Run 4	0.75	277	1.022	315.8	328.8	51.76
Run 5	0.8	277	1.022	315.8	325.5	48.52
Run 6	0.85	277	1.022	315.8	322.7	45.67
Run 7	0.9	277	1.022	315.8	320.2	43.13

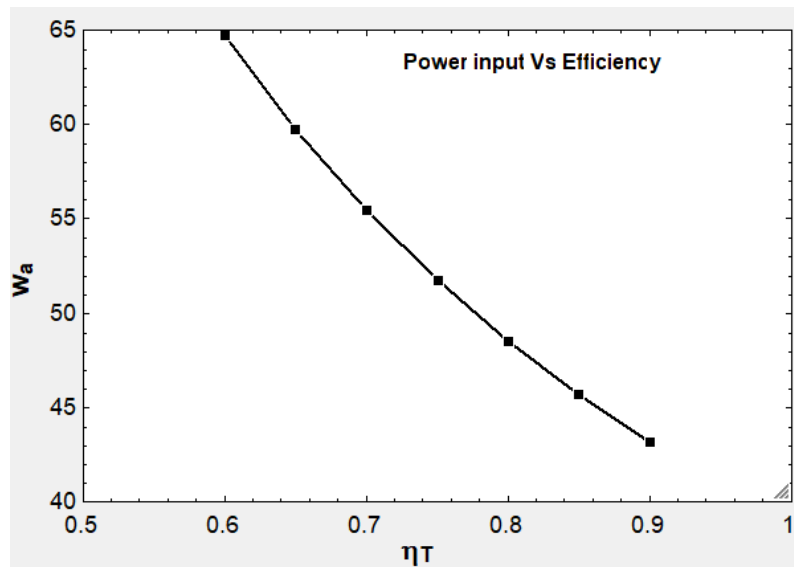
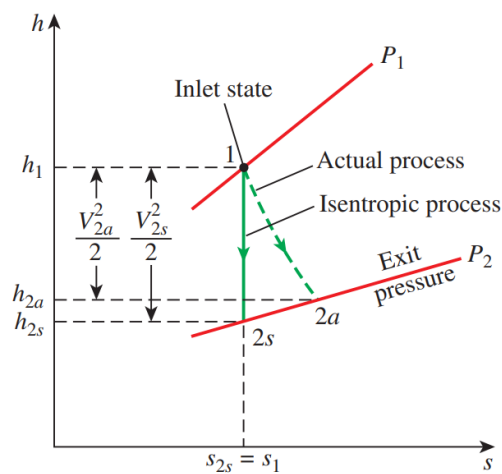


Figure 4: Power input vs efficiency plot for compressor

4.3 Nozzle

General definition of isentropic efficiency for nozzle when potential changes across nozzle and inlet velocity is small compared to outlet velocity.

Figure 5: The h - s diagram of the actual and isentropic process of a nozzle

$$\eta_N = \frac{\text{Actual KE at nozzle exit}}{\text{Isentropic KE at nozzle exit}} = \frac{V_{2a}^2}{V_{2s}^2} = \frac{h_1 - h_{2a}}{h_1 - h_{2s}} \dots \dots \dots eq(5)$$

Isentropic exit speed

$$V_{2s} = \sqrt{2(h_1 - h_{2s})} \dots \dots \dots eq(6)$$

Actual exit speed

$$V_{2a} = \sqrt{2(h_1 - h_{2a})} \dots \dots \dots eq(7)$$

4.3.1 Manual Calculations

Sample calculation for isentropic efficiency of 60% and given inlet and outlet conditions.

Considering air as ideal gas. Properties of air were as follows.

$$R = 0.287 \text{ kJ/kg.K}$$

$$c_p = 1.005 \text{ kJ/kg.K}$$

$$\gamma = \text{specific heat ratio} = 1.4$$

From ideal gas

$$h_1 = c_p T_1 = 1.005 * 950 = 954.75 \text{ kJ/kg}$$

For isentropic process of ideal gas temperature and pressure relation is [2].

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

From above equation for isentropic process

$$T_{2s} = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = 950 \left(\frac{110}{200}\right)^{\frac{1.4-1}{1.4}} = 800.83 \text{ K}$$

$$h_{2s} = 1.005 * 800.83 = 804.84 \text{ kJ/kg}$$

So, from definition of isentropic efficiency i.e. eq (5)

$$0.6 = \frac{954.75 - h_{2a}}{954.75 - 800.83}$$

Solve above equation for h_{2a}

$$h_{2a} = 862.4 \text{ kJ/kg}$$

Isentropic exit speed from eq (6)

$$V_{2s} = \sqrt{2(954.75 - 804.84) * 1000} = 547.6 \text{ m/s}$$

Actual exit speed from eq (7)

$$V_{2a} = \sqrt{2(954.75 - 862.4) \cdot 1000} = 429.8 \text{ m/s}$$

Following table show manual calculation results at different efficiencies.

Table 6: Exit speed for different nozzle isentropic efficiencies

Isentropic Efficiency	h_1	T_{2s}	h_{2s}	h_{2a}	Actual exit Speed
%	kJ/kg	K	kJ/kg	kJ/kg	kJ/kg
60	954.75	800.83	804.84	862.4	429.8
65	954.75	800.83	804.84	854.7	447.3
70	954.75	800.83	804.84	847.0	464.2
75	954.75	800.83	804.84	839.3	480.5
80	954.75	800.83	804.84	831.6	496.3
85	954.75	800.83	804.84	823.9	511.6
90	954.75	800.83	804.84	816.2	526.4

4.3.2 EES Calculations

"inlet consitions" P_1= 200 [kPa] T_1=677 [C] m_dot=3 [kg/s] "outlet pressure" P_2= 110 [kPa] h_1= Enthalpy(Air,T=T_1) s_1=Entropy(Air,T=T_1,P=P_1) T_2=Temperature(Air,s=s_1,P=P_2) h_2s=Enthalpy(Air,T=T_2) "Isentropic Efficiency " "eta_T= 0.6" h_2a=h_1 - eta_T*(h_1-h_2s) "Isentropic exit speed" V_2s= sqrt(2*(h_1-h_2s)*1000) "Actual exit speed" V_2a= sqrt(2*(h_1-h_2a)*1000) "Exit momentum for isentropic speed" p_exit_s=m_dot * V_2s "Exit momentum for isentropic speed" p_exit_a=m_dot * V_2a	Formatted Equations inlet consitions $P_1 = 200 \text{ [kPa]}$ $T_1 = 677 \text{ [C]}$ $\dot{m} = 3 \text{ [kg/s]}$ outlet pressure $P_2 = 110 \text{ [kPa]}$ $h_1 = h(\text{Air}, T=T_1)$ $s_1 = s(\text{Air}, T=T_1, P=P_1)$ $T_2 = T(\text{Air}, s=s_1, P=P_2)$ $h_{2s} = h(\text{Air}, T=T_2)$ Isentropic Efficiency $\eta_T = 0.6$ $h_{2a} = h_1 - \eta_T \cdot (h_1 - h_{2s})$ Isentropic exit speed $V_{2s} = \sqrt{2 \cdot (h_1 - h_{2s}) \cdot 1000}$ Actual exit speed $V_{2a} = \sqrt{2 \cdot (h_1 - h_{2a}) \cdot 1000}$ Exit momentum for isentropic speed $p_{\text{exit},s} = \dot{m} \cdot V_{2s}$ Exit momentum for isentropic speed $p_{\text{exit},a} = \dot{m} \cdot V_{2a}$
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Table 7: Nozzle exit speeds from EES

	1	2	3	4	5	6	7	8	9	10	
	η_N	h_1	s_1	T_2	h_{2s}	h_{2a}	V_{2s}	V_{2a}	$p_{exit,s}$	$p_{exit,a}$	
Run 1	0.6	989.7	6.714	541.6	838.5	899	549.9	425.9	1650	1278	
Run 2	0.65	989.7	6.714	541.6	838.5	891.4	549.9	443.3	1650	1330	
Run 3	0.7	989.7	6.714	541.6	838.5	883.8	549.9	460.1	1650	1380	
Run 4	0.75	989.7	6.714	541.6	838.5	876.3	549.9	476.2	1650	1429	
Run 5	0.8	989.7	6.714	541.6	838.5	868.7	549.9	491.8	1650	1476	
Run 6	0.85	989.7	6.714	541.6	838.5	861.2	549.9	507	1650	1521	
Run 7	0.9	989.7	6.714	541.6	838.5	853.6	549.9	521.7	1650	1565	

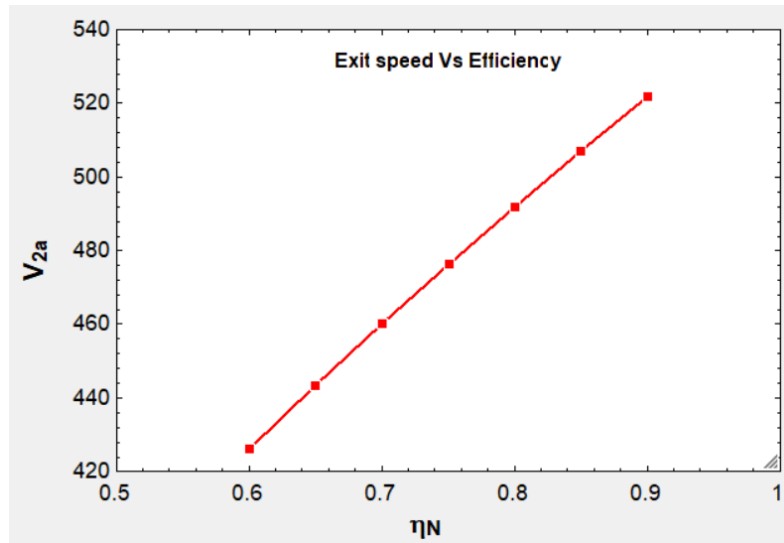


Figure 6: Exit speed plot vs Efficiency

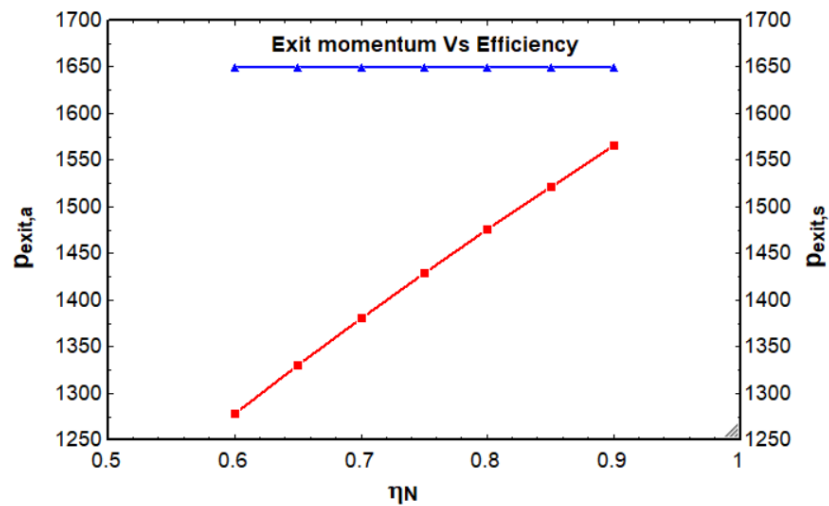


Figure 7: Exit momentum vs Efficiency

5 Results and Discussions

This effort entailed a comprehensive examination of three fundamental steady flow devices, namely turbines, compressors, and nozzles. Every individual device was powered using a distinct fluid, namely steam for the turbine, Refrigerant 134a for the compressor, and air for the nozzle. The main aim of this study was to investigate the impact of isentropic efficiency on the power output of the turbine, power input of the compressor, and exit velocity of the nozzle.

To encompass a wide range of operating conditions, a set of seven isentropic efficiency values were meticulously selected for each application. Assumptions were made regarding the inlet circumstances and exit pressures, thereby establishing a basis for a methodical investigation. The turbine's power output, the compressor's power input, and the nozzle's exit speed were subsequently graphed in relation to the chosen isentropic efficiency values.

The findings demonstrated a distinct association between isentropic efficiency and the operational effectiveness of the steady flow devices. In the context of the turbine, an augmentation in isentropic efficiency was seen to be associated with a significant elevation in power generation. On the other hand, an increase in isentropic efficiency for the compressor led to a reduction in power input. The exit speed of the nozzle had a positive correlation with the isentropic efficiency, indicating that higher efficiency resulted in greater speed at the nozzle exit. The aforementioned results highlight the importance of isentropic efficiency in assessing the efficacy of these devices.

In order to enhance the depth of the analysis, additional factors were taken into account, namely the mass flow rate and exit momentum, for both actual and isentropic scenarios. The extensive methodology employed in this study facilitated a comprehensive comprehension of the performance indicators, hence illuminating the subtle dynamics inherent in these devices.

The analysis was performed utilising both the EES software and hand computations, so assuring a comprehensive investigation of the findings. The consistency and validity of the results obtained from both methods serve to authenticate the accuracy of the findings. The utilisation of a dual-method approach in this study serves to bolster its overall strength and engender a sense of trustworthiness in the obtained results.

6 Conclusions

In summary, the experiment underscores the significance of isentropic efficiency in dictating the operational effectiveness of steady flow devices. The patterns that have been noticed highlight the importance of achieving higher isentropic efficiency in order to improve power

output, decrease power input, and increase exit speed. The aforementioned insights provide essential recommendations for optimising turbines, compressors, and nozzles in real engineering applications, thereby contributing to the broader debate in the fields of fluid dynamics and thermodynamics.

7 References

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- [2] Çengel, Y.A., Boles, M.A. and Kanoğlu, M. (2024) Thermodynamics: An engineering approach. New York, NY: McGraw Hill.